

Ch3: Data Modeling Using the Entity-Relationship (ER) Model

Textbook: Elmasri and Navathe, *Fundamentals of Database Systems*, 7th Edition, Chapter 3. Read Sections 3.1-3.7, 3.9.1-3.9.2, and 3.10.

Teaching Summary

An ER diagram describes the objects a database represents, their properties, and the rules connecting them. Start with a small campus example, draw one concept at a time, and then assemble the design. The tables below are descriptions of fictional objects and facts, not proposed relational tables or SQL output.

Ch1 introduced databases and their users. Ch2 distinguished a schema from its current state. Here you will design a conceptual schema from stated requirements. You need no SQL, Python, drawing application, or database installation. Paper is enough for the drawing exercises.

By the end, you should be able to identify entities and attributes, justify keys, read and draw relationship constraints, identify weak entities, and explain why a three-part fact may need a ternary relationship. You should also be able to check whether a diagram permits a state that the requirements forbid.

All campus records are synthetic. The main example covers **Fall 2026 only**. A section is a particular class offering of a course during that term. Separate small variants introduce rooms, cards, and mentoring; they are not additional requirements for the final campus diagram.

1. From Requirements to a Diagram

An **entity-relationship (ER) model** is a high-level conceptual data model. It describes meaning without choosing relational tables or disk structures. A **miniworld** is the portion of the world that the database represents.

Consider two requirements: each section belongs to exactly one course, and different courses may reuse a section number. The first requirement introduces COURSE, SECTION, and a connection. The second asks how a section is identified. It does not tell us how to implement that identification in SQL.

The core campus requirements are:

Fact or rule	Meaning in this example
Students have unique StudentId values and zero or more recorded phone numbers.	Different students cannot share a StudentId.
Courses have unique CourseCode values.	A course can exist before any section is offered.
Each section belongs to exactly one course.	SectionNo is unique within its course, not across all courses.
Each section has exactly one instructor.	An instructor may teach zero or more sections.
Students may enroll in several sections.	A section may have zero or more students.
A grade, when available, belongs to one enrollment.	A student's different enrollments may have different grades.
An instructor may approve a student for a particular course.	Each approval names all three participants.

InstructorId uniquely identifies an instructor. Approvals are recorded separately from enrollment: these requirements do not make an approval a prerequisite for enrollment. Repeated approval events with the same three participants are outside this example.

Prediction: Which of the first two course-section requirements would be lost if a drawing showed only two named boxes without a relationship or constraints?

Start with facts and rules

Start with facts and rules

Original teaching illustration | Read with the worked example

Requirement

Given statement	Design question
Each section belongs to one course.	Which objects are related?
Section numbers repeat across courses.	What identifies a section?

Conceptual decision

Object or rule	ER representation
COURSE and SECTION	Entity types
One owner per section	Relationship and constraints

An ER diagram describes the miniworld before relational mapping or storage choices.

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Read the Diagram

The boxes alone name the types but do not express which course owns a section. Adding a relationship states the connection. Adding constraints states how many owners each section may have. Identification still needs a key analysis.

A requirement such as "show all sections taught by I1" is a useful design check: the schema must preserve enough information to answer it. Choosing an index is a later physical-design question, not an ER symbol.

Practice: For "each library copy belongs to one book edition," name the two types and write one question about copy identification. Check that your question distinguishes a copy number unique everywhere from one unique only within an edition.

2. Entity, Entity Type, and Entity Set

An **entity** is one distinguishable object, such as the student identified by S101. An **entity type** describes entities with the same kinds of properties. An **entity set** contains the entities of that type at a particular time.

STUDENT is the type. S101, S102, and S103 are three entities in its current set. StudentId is an attribute; S101 is an attribute value used here to identify one entity. The person is not the same thing as the identifier string.

Prediction: Adding S104 without changing any attribute definitions changes the entity type, the current entity set, or both?

One entity is not the whole entity type

One entity is not the whole entity type

Original teaching illustration | Read with the worked example

Type description

Type	Properties
STUDENT	StudentId, Name, Phone

Current illustrative set

StudentId	Name
S101	Alex Lin
S102	Blair Wu
S103	Casey Chen

Adding S104 changes the set, not the type definition.

STUDENT is the type; S101 is one entity; these three students form the displayed current set.

STUDENT is the type; S101 is one entity; these three students form the displayed current set.

Read the Diagram

The displayed set grows from three students to four. The type still describes students using the same properties. This is the Ch2 distinction between the schema and a state, now applied to entities.

A rectangle in an ER schema names an entity type, not one particular student. Do not draw a new schema rectangle for every row in an example.

Practice: Describe COURSE at all three levels: one course, the type, and a current set of two courses. Check that the set contains objects, whereas the type description states their shared properties.

3. Attributes Describe an Entity

An **attribute** describes a property of an entity. For this expanded student profile, suppose the application needs to search given and family names separately, keep several phone numbers, and count current enrollments. Draw an attribute as an oval connected to its entity type. This connection attaches the property; it is not a separate relationship type.

Student	GivenName	FamilyName	Recorded Phone values	EnrollmentCount
S101	Alex	Lin	02-0000-0101; 02-0000-0102	2
S102	Blair	Wu	02-0000-0201	1
S103	Casey	Chen	None recorded	0

The phone strings are invented. Semicolons separate values for reading; they are not a recommendation to store a list in one SQL column.

A **simple attribute** has no components used by this model. A **composite attribute** groups meaningful components: Name contains GivenName and FamilyName. Name need not be composite in every application; that choice depends on the information the application needs.

A **single-valued attribute** has one value for an entity when known. A **multivalued attribute** may have several values for one entity. Phone is multivalued here. Composite and multivalued describe different properties. For example, an optional Contact attribute could contain several (Label, PhoneNumber) pairs. Each pair has components, and a student can have several pairs. This is a small example of a **complex attribute**.

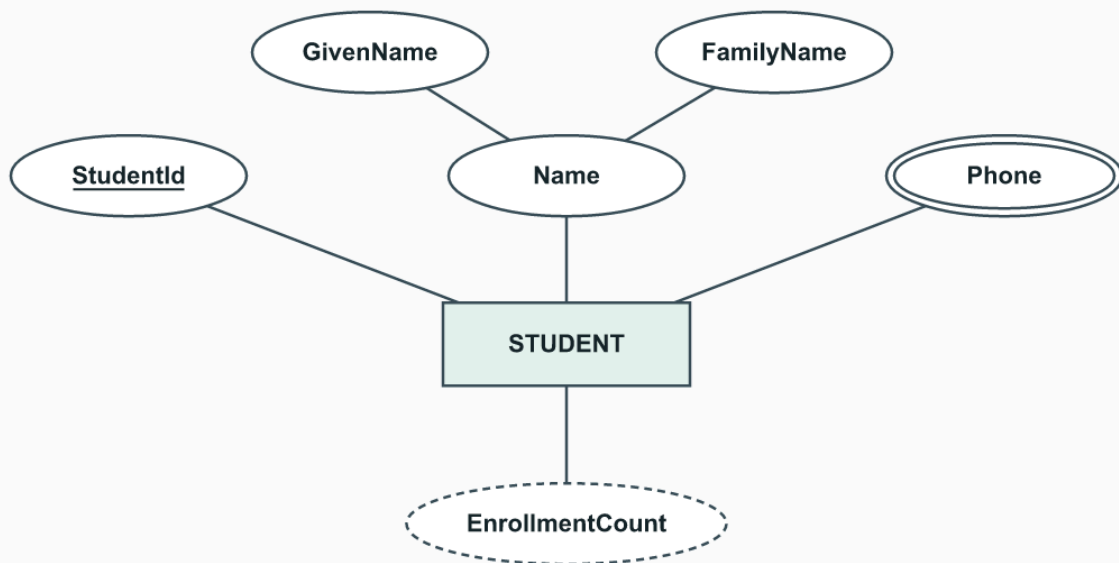
A **stored attribute** supplies a recorded value. A **derived attribute** is calculated from other information. EnrollmentCount is derived by counting the student's enrollment relationships; it is not the sum of grade values.

Prediction: Which attribute needs a double oval, and which needs a dashed oval? How many phone values does S101 have?

Different attribute properties use different symbols

Different attribute properties use different symbols

Original synthetic campus example



Name has components; Phone can have several values; EnrollmentCount is calculated from enrollments.

Name has components; Phone can have several values; EnrollmentCount is calculated from enrollments.

Read the Diagram

Phone has a double oval because S101 may have more than one phone value. EnrollmentCount has a dashed oval because it is calculated. Name connects to two component ovals. StudentId is underlined because it is a key, discussed next.

Count S101's two enrollment facts in Section 6: the derived value is 2. If one enrollment is removed, this count becomes 1. Its value depends on the current relationship set, not on the number of phones.

Practice: Draw Contact for two invented (Label, PhoneNumber) pairs. Then change the requirements so that only one phone may be recorded per student. Check which multiplicity symbol changes; composite structure and multiplicity must not be treated as the same property. This profile is an explanatory variant; the final core diagram retains only StudentId and Phone.

4. Keys Identify Entities

A **key attribute** has a value that uniquely identifies every entity in each legal state of the entity set. A key can be composite. Its components must be needed together; adding an unnecessary component does not produce a minimal key.

Uniqueness in a few sample records is not enough to establish a key. Two students may have different names today, but the requirements allow future students to share a name. StudentId is a key because the rule requires uniqueness.

For a separate campus-room example, assume a room number is unique only within its building:

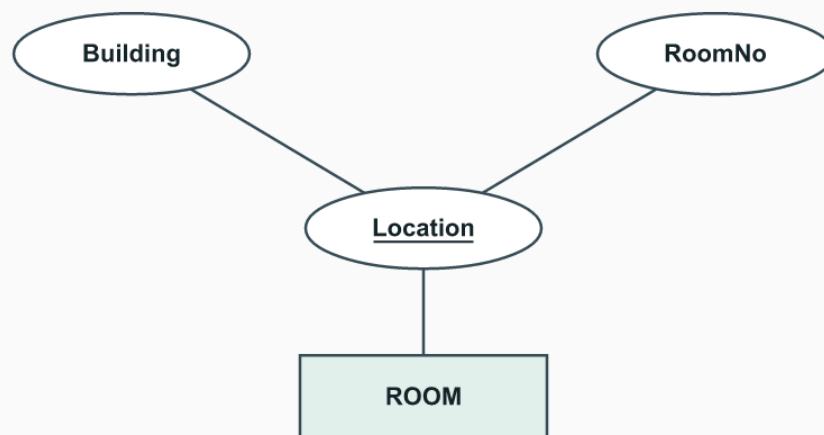
Building	RoomNo
A	101
A	102
B	101

Prediction: Does RoomNo alone identify a room? Does Building alone? Which combination distinguishes all rooms under the stated rule?

One composite key is not two separate keys

One composite key is not two separate keys

Original synthetic campus example



Underline Location, not Building and RoomNo separately. Neither component identifies a room alone.

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Read the Diagram

RoomNo 101 occurs twice; Building A also occurs twice. The composite Location(Building, RoomNo) identifies a room, and removing either component loses that guarantee. Underline the composite Location oval.

In the book's ER notation, separately underlining Building and RoomNo would claim that each is a key independently. That would be wrong here. An entity type may have more than one independent key if the requirements guarantee each. This ER presentation does not select a "primary key"; that choice belongs to the relational model and mapping chapters.

Practice: Suppose rooms receive a new RoomId unique across all buildings, while the Location rule remains. Draw both valid keys. Check that you have not turned two independent keys into one larger composite key.

5. Domains and Missing Values

A **value set**, or **domain**, specifies allowed values of an attribute. It is not simply the values currently present.

For an expanded COURSE description, assume Credits must be a whole number from 1 through 6. The two observed courses currently have 2 and 3 credits. A proposed value of 5 is still legal. A value of 2.5 fails the whole-number condition, and 7 exceeds the limit. These are example rules, not university policy.

Missing information also needs a meaning. The book uses **NULL** for information that is unknown or not applicable. A grade may be unknown because marking is unfinished. In a separate staff profile, ParkingPermitNo may be inapplicable because the person has no permit. Not knowing whether a permit exists is different from knowing that none exists.

Prediction: Is 5 rejected because it is absent from today's sample? Can an unrecorded grade automatically be interpreted as zero?

Allowed values are not just the observed values

Allowed values are not just the observed values

Original teaching illustration | Read with the worked example

Course rule

Attribute	Allowed values
Credits	Whole numbers from 1 through 6

Check proposed values

Value	Decision
5	Allowed
2.5	Reject: not whole
7	Reject: above 6

Under this example's rule, 5 is allowed even though only 2 and 3 occur in the current courses.

Under this example's rule, 5 is allowed even though only 2 and 3 occur in the current courses.

Read the Diagram

Five satisfies the declared domain. The displayed rejected values violate specific conditions, not merely an observed pattern. An unrecorded grade is not evidence of a score of zero.

"No phone recorded" likewise does not prove that a student owns no phone. A conceptual explanation can distinguish reasons for missing information. The later SQL chapter will discuss what SQL NULL operations actually do.

Practice: Check proposed Credits values 1, 6, 0, and 3.5 against both conditions. For an absent advisor name, write two different possible reasons and one question that distinguishes them. Keep the reason separate from a claim about how a DBMS stores it.

6. Relationships Connect Entities

A **relationship instance** associates particular participating entities. An enrollment connecting S101 to the section DB101 / 1 is one instance. **ENROLLS_IN** is the relationship type; its current **relationship set** contains the enrollment instances currently recorded. Draw a relationship type as a diamond connected to its participating entity types. These lines are undirected connections, not a sequence of actions.

The slash notation DB101 / 1 identifies a section by course and section number for human reading. It is not a foreign-key declaration. Section 12 explains why this identity needs both parts.

Student	Section	Grade
S101	DB101 / 1	80
S101	CS102 / 1	90
S102	DB101 / 1	70

The **degree** counts participating roles. This relationship is binary because each enrollment connects a student and a section. The number of attributes or the number of current enrollment facts does not determine its degree.

Prediction: How many enrollment instances are shown? How many involve S101? Does S103's absence mean that S103 must be deleted from STUDENT?

An enrollment connects a student to a section

An enrollment connects a student to a section

Original teaching illustration | Read with the worked example

ENROLLS_IN instances

Student	Section (course / number)
S101	DB101 / 1
S101	CS102 / 1
S102	DB101 / 1

Count participation

Student	Enrollment count
S101	2
S102	1
S103	0

The three rows describe three relationship instances, not three relationship types.

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Read the Diagram

There are three enrollment instances; two involve S101 and none involves S103. The requirements allow students with no enrollment, so S103 may remain. The relationship set is separate from the entity set.

Practice: Add an enrollment linking S103 to DB101 / 2, with the grade not yet known. Recount instances and each student's participation. Check that you added a relationship instance, not a new student entity or relationship type.

7. Maximum Cardinality

The **cardinality ratio** of a binary relationship restricts its maximum participation pattern. Read the requirements in both directions.

Relationship	Rule for one object on the left	Rule for one object on the right
STUDENT HOLDS CARD, a separate variant	At most one card	At most one student
INSTRUCTOR TEACHES SECTION	May teach several sections	At most one instructor
STUDENT ENROLLS_IN SECTION	May enroll in several sections	May contain several students

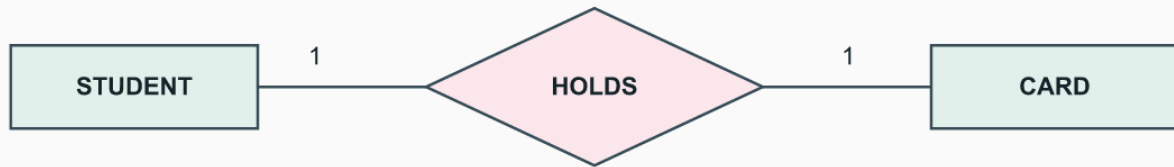
These give 1:1, 1:N, and M:N respectively. M and N mean that the diagram gives no fixed numeric upper bound on that side; they do not require every entity to participate many times.

Prediction: I1 teaches DB101 / 1 and DB101 / 2. Which maximum rule would be violated if I2 also taught DB101 / 1 in this example?

Compare maximum counts in both directions

Compare maximum counts in both directions

Original synthetic campus example



At most one card per student; at most one student per card.



One instructor may teach many sections; one section has at most one instructor.



One student may enroll in many sections; one section may have many students.

These panels isolate maximum ratios. Participation requirements are added in the next figure.

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Read the Diagram

One section would have two instructors, violating its maximum of one. If teaching two sections does not violate the rule.

In the book's maximum-ratio notation, the 1 near INSTRUCTOR means that one SECTION has at most one instructor. The N near SECTION permits an instructor to teach many sections. Follow the whole relationship rather than reading the nearest numeral as that entity's own participation count. Section 9 introduces a different placement convention.

The three panels isolate the maximum counts. Participation requirements are added next. Relationship close-ups omit keys and weak-entity markings until Section 12 explains identification; Section 15 combines the symbols. A small state with one instructor and one section does not prove the relationship is 1:1; permitted future states matter.

Practice: For a team-teaching variant, allow several instructors per section while each instructor may still teach several sections. Redraw only the maximum ratio. Check your drawing using two instructors connected to the same section.

8. Total and Partial Participation

A **participation constraint** specifies whether every entity must participate. **Total participation** requires at least one relationship instance for each entity of the participating type. **Partial participation** permits zero.

The final TEACHES rule is stronger than "at most one": each section must have exactly one instructor. SECTION therefore has total participation. INSTRUCTOR has partial participation because an instructor can teach no section.

Prediction: Is a new instructor I3 with no teaching assignment allowed? Is a new section with no instructor allowed?

A double line requires participation

A double line requires participation

Original synthetic campus example



Single line: zero is allowed.

Double line: at least one is required.

Every SECTION must have a teacher. An INSTRUCTOR may currently teach no section.

Every SECTION must have a teacher. An INSTRUCTOR may currently teach no section.

Read the Diagram

I3 is allowed; the unassigned section is not. The double line from SECTION to TEACHES expresses the minimum of one. The maximum ratio supplies the upper bound of one instructor per section. These are different constraints.

Partial participation does not require an unassigned instructor to exist in every current state. It merely permits one. Total participation does not by itself make an entity weak: a CARD with its own globally unique CardNo can require an owner without depending on the owner's key for identification.

Practice: Change the rule to require every instructor to teach at least one section, without changing the maximum counts. Mark the line that changes. Then decide whether adding I3 alone would still be legal and cite the new rule.

9. Read Min-Max Labels Carefully

The book also presents **(min, max)** notation. A pair beside an entity type counts how many relationship instances **one entity of that type** participates in.

For TEACHES, put (0,N) beside INSTRUCTOR and (1,1) beside SECTION. This represents the same requirements as the preceding complete TEACHES diagram. The pair supplies both minimum and maximum, so this figure uses single connection lines instead of the earlier single/double participation convention.

Prediction: Does (1,1) beside SECTION mean one student per section or one TEACHES instance per section?

Place this notation beside the entity being counted

Place this notation beside the entity being counted

Original synthetic campus example



Min-max replaces the maximum-ratio and single/double-line convention here.

Here (0,N) counts sections for one instructor; (1,1) counts teachers for one section.

Here (0,N) counts sections for one instructor; (1,1) counts teachers for one section.

Read the Diagram

It means exactly one TEACHES instance per section: one instructor. It says nothing about ENROLLS_IN or the number of students.

Do not copy the positions of 1 and N from the maximum-ratio diagram into min-max pairs. The two notations place their counts differently. UML also has its own placement convention; do not silently mix it into this ER diagram.

Practice: An instructor must teach at least one and at most three sections. Write the new pair beside INSTRUCTOR and test counts 0, 1, 3, and 4. Keep SECTION's rule unchanged. Your check should name both accepted and rejected counts.

10. Recursive Relationships and Roles

A **recursive relationship** connects entities of the same type in different **roles**. For a separate mentoring variant, one student may mentor several students, and a student may have at most one mentor. Both roles permit zero.

Mentor	Mentee
S101	S102
S101	S103

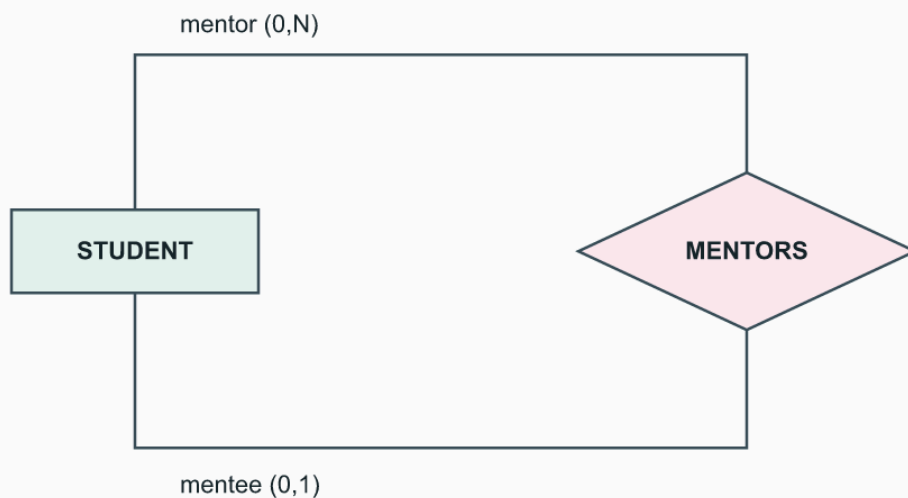
MENTORS is binary even though only one entity-type rectangle is needed. Each instance has a mentor participation and a mentee participation.

Prediction: Could the same two names in reversed order describe a different mentoring fact? Which role has two participations in this sample?

One entity type can participate twice

One entity type can participate twice

Original synthetic campus example



MENTORS is binary: one student plays mentor and another plays mentee. The roles distinguish the two.

MENTORS is binary: one student plays mentor and another plays mentee. The roles distinguish the two.

Read the Diagram

S101 is the mentor in two instances. Reversing the names reverses the roles; S102 mentoring S101 is not the same fact as S101 mentoring S102. The two lines connect the same type to different roles of the same relationship, not two different student types.

If the application forbids self-mentoring or cycles, write those as additional rules. The displayed min-max constraints alone do not prohibit either.

Practice: Add S102 as mentor of S103 and inspect S103's mentee participation count. Separately consider S101 mentoring S101. Identify which question can be answered by the displayed counts and which needs an additional business rule.

11. Attributes of a Relationship

An attribute can describe a relationship instance rather than one participating entity. Grade belongs to an enrollment: it needs both the student and the section.

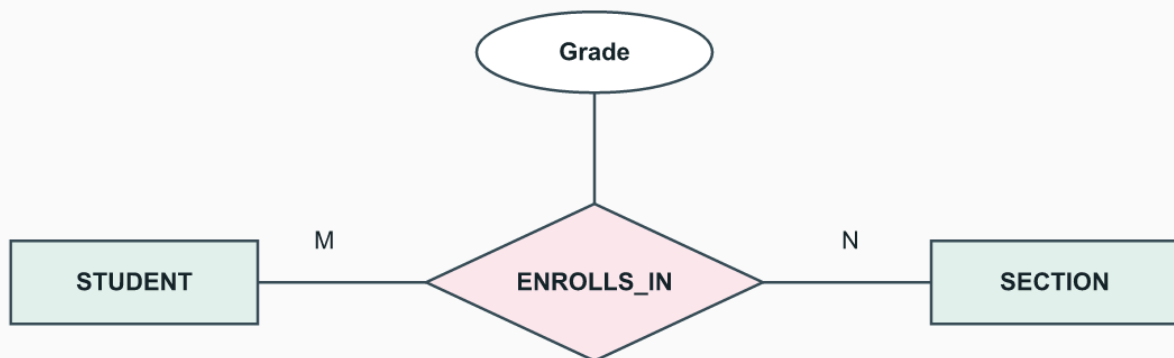
Student	Section	Grade
S101	DB101 / 1	80
S101	CS102 / 1	90
S102	DB101 / 1	70

Prediction: Can one Grade value attached to S101 represent both of S101's grades? Can one Grade attached to DB101 / 1 represent both of that section's grades?

A grade belongs to a particular enrollment

A grade belongs to a particular enrollment

Original synthetic campus example



S101 has 80 in DB101 / 1 and 90 in CS102 / 1. One grade on STUDENT cannot describe both.

S101 has 80 in DB101 / 1 and 90 in CS102 / 1. One grade on STUDENT cannot describe both.

Read the Diagram

Neither single value is enough. S101 has 80 and 90; DB101 / 1 has grades 80 and 70 for different students. The Grade oval connects to ENROLLS_IN.

The argument depends on these M:N facts. It does not establish that every relationship attribute must always remain on a relationship: under suitable 1:N or 1:1 constraints, an attribute can be placed on an entity without losing which relationship instance it describes.

Practice: Suppose each enrollment records a submission status, such as "submitted" or "not submitted." Place the attribute and construct two facts showing why attaching it to STUDENT alone could lose information. Check that your two facts involve the same student and different sections.

12. Weak Entities and Partial Keys

A **weak entity type** lacks a complete identifying key of its own. An **owner entity type** supplies part of its identification through an **identifying relationship**. A **partial key** distinguishes weak entities belonging to the same owner or owner combination. Weak types normally have a partial key; a later example explains when the owners alone suffice.

For this one-term example, SECTION has no globally unique SectionId:

Owning COURSE	SectionNo	Section identified
DB101	1	DB101 / 1
DB101	2	DB101 / 2
CS102	1	CS102 / 1

Prediction: Are the two occurrences of SectionNo 1 an error? Would a second section numbered 1 under DB101 be allowed?

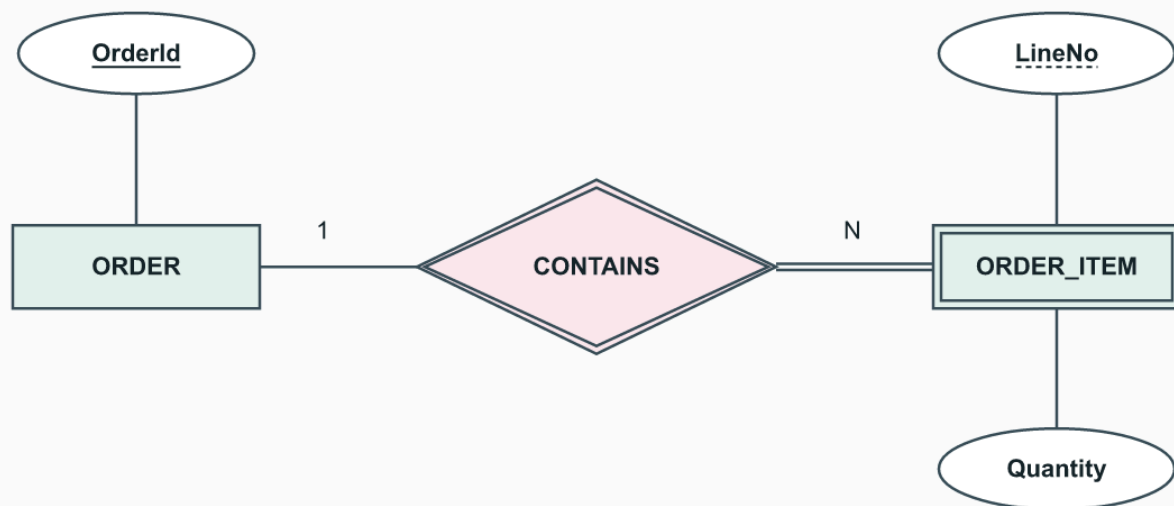
OrderId	LineNo	Product	Quantity
O10	1	Notebook	2
O10	2	Pen	3
O20	1	Notebook	2

Prediction: Are O10 / 1 and O20 / 1 the same item because their product, quantity, and line number match?

An item number identifies an item only within its order

An item number identifies an item only within its order

Original synthetic campus example



O10 / 1 and O20 / 1 are distinct items. Quantity describes an item but is not its partial key.

O10 / 1 and O20 / 1 are distinct items. Quantity describes an item but is not its partial key.

Read the Order Item Diagram

They are different order items because their owners differ. Follow three steps: identify the order, find LineNo within that order, then read the item's properties. The full identifying information is (OrderId, LineNo). LineNo is only a partial key. A fourth item with O10 / 1 would violate the stated identification rule.

The double rectangle marks ORDER_ITEM; the double diamond marks CONTAINS. Every item must participate in CONTAINS, but an order may have no items while being prepared. That optional order participation is an explicit example rule.

Practice: Add O20 / 2, then try adding another O10 / 2 with a different quantity. Explain why changing Quantity cannot repair a duplicate identity.

12.2. A Required Owner Does Not Automatically Make an Entity Weak

Now consider campus access cards. CardId is unique across all issued cards. Each card must belong to exactly one student; a student may have zero or several cards. This is a separate variant from the earlier one-card maximum example.

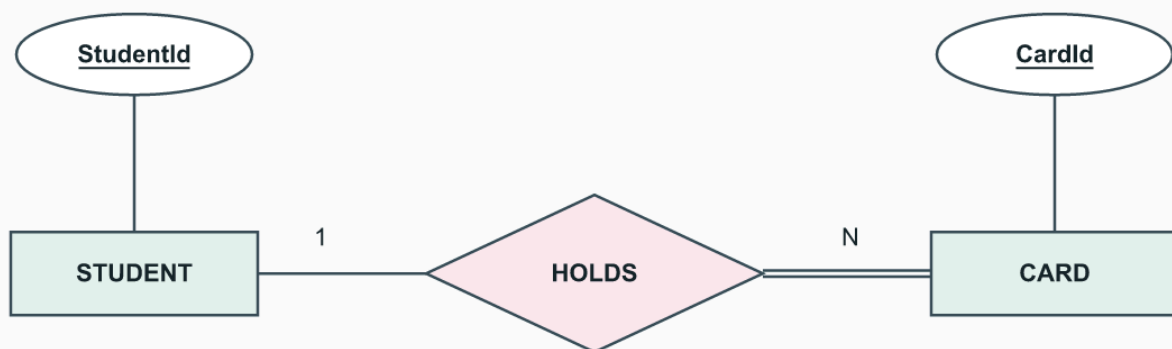
CardId	StudentId
K10	S101
K11	S101
K12	S102

Prediction: Is CARD weak merely because a card without an owner is forbidden?

CARD has its own key even when ownership is required

CARD has its own key even when ownership is required

Original synthetic campus example



CardId is globally unique. A required owner changes participation, not this independent identification.

CardId is globally unique. A required owner changes participation, not this independent identification.

Read the Card Diagram

CARD is a regular, or strong, entity type: CardId identifies a card independently. Use a single rectangle and an ordinary relationship diamond. The double participation line still applies to CARD because ownership is required.

Compare K10 with order item O10 / 1. K10 is already globally identifying. The value 1 alone cannot identify an order item. The decisive question is identification, not whether one object depends on another object's existence.

Practice: Remove the global CardId rule and number cards only within each student. State the new identifying combination and which symbols must change.

12.3. A Weak Entity Can Own Another Weak Entity

Extend the order example to record notes about individual items. NoteNo is unique within one order item; it may repeat for another item. ITEM_NOTE has no global ID.

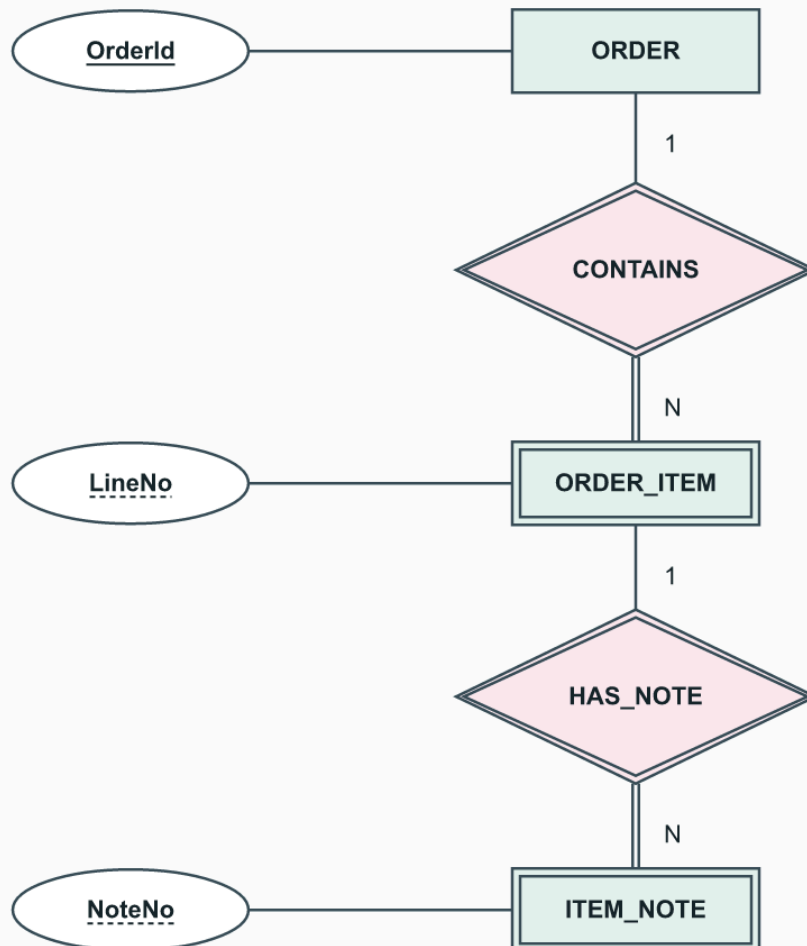
OrderId	LineNo	NoteNo	Note text
O10	1	1	Gift wrap
O10	2	1	Blue ink
O20	1	1	No wrap

Prediction: Is (LineNo, NoteNo) enough to identify the first and third notes?

Follow identification through every owner

Follow identification through every owner

Original synthetic campus example



A note is identified by OrderId, LineNo, and NoteNo. ORDER_ITEM is both weak and an owner.

A note is identified by OrderId, LineNo, and NoteNo. ORDER_ITEM is both weak and an owner.

Read the Nested Owner Diagram

No: both have (1,1). Identify ORDER with OrderId, then ORDER_ITEM with its owner's identity plus LineNo, then ITEM_NOTE with that item's identity plus NoteNo. The complete combination is (OrderId, LineNo, NoteNo).

ORDER_ITEM is both a weak entity and an owner of ITEM_NOTE. Both weak types participate totally in their own identifying relationships. A parent item can have zero notes; requiring a note's owner does not require every item to have notes.

Practice: Add note 2 to O10 / 1. Compare its identity with note 2 on O20 / 1. Check every level rather than looking only at the nearest partial key.

12.4. Two Owners Can Be Needed Together

For fictional internship interviews, STUDENT has StudentId and COMPANY has CompanyId. A student may interview at several companies, and a company may interview several students. VisitNo numbers interviews within one student-company pair.

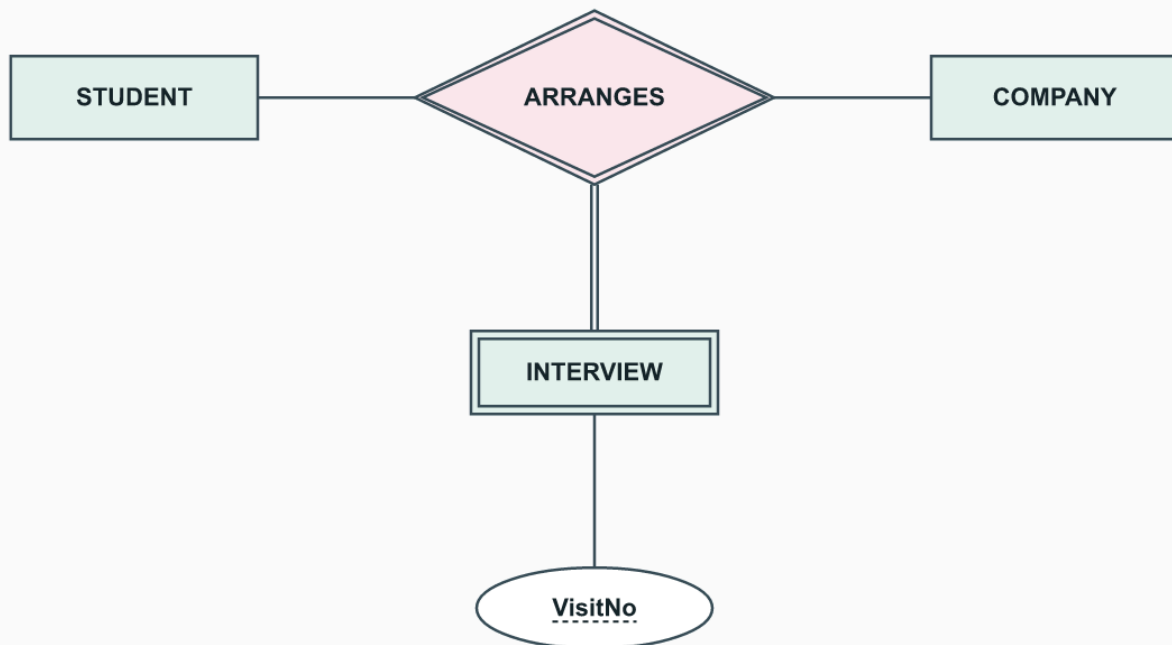
StudentId	CompanyId	VisitNo
S101	C1	1
S101	C2	1
S102	C1	1

Prediction: Is (StudentId, VisitNo) enough? Is (CompanyId, VisitNo) enough?

An interview number is local to a student-company pair

An interview number is local to a student-company pair

Original synthetic campus example



Fix both owners, then use VisitNo. StudentId or CompanyId plus VisitNo alone is insufficient.

Fix both owners, then use VisitNo. StudentId or CompanyId plus VisitNo alone is insufficient.

Read the Two-Owner Diagram

Neither works: the first combination repeats across the first two rows and the second repeats across the first and third. The complete identity is (StudentId, CompanyId, VisitNo). INTERVIEW is weak and has two owner types. VisitNo distinguishes interviews only after both owners have been fixed.

The identifying relationship has three participants: STUDENT, COMPANY, and INTERVIEW. The double line is on INTERVIEW's participation. This is an original illustration of the multiple-owner idea in Section 3.9.1, not a copy of its job-offer example.

Practice: Add a second visit for S101 at C1. Explain why another first visit for S101 at C1 would be a duplicate, even if its interview location changed.

A weak type normally has a partial key, but not every representation needs one. If exactly one interview were allowed for each student-company pair, those two owners alone could identify it. In that alternative design, VisitNo is unnecessary. If repeated visits are allowed, the pair alone is insufficient.

Choosing between an Attribute and a Weak Entity

Suppose only a list of phone labels and numbers is needed for each student. A multivalued composite Contact attribute may be enough. If each contact must also participate in a verification relationship with a staff member, representing CONTACT as an entity lets that independent relationship be drawn explicitly. With a label unique only within its student, CONTACT can be weak.

Prediction: Does being multivalued alone force Contact to be a weak entity?

Check the Design Choice

No. Both representations can describe several contacts. The separate verification relationship is a concrete reason to choose an entity representation. For example, S101's "home" contact and S102's "home" contact remain distinct through their owners. If no independent contact relationship is needed, the complex attribute may be sufficient. This is a design choice based on requirements.

Practice: Give CONTACT a globally unique ContactId instead. Explain why it can then be regular even though every contact still belongs to a student.

13. Refine a Design from Its Requirements

Design is iterative. An initial list may describe "teacher name" as a section attribute. If the application also needs instructor identity, name, and teaching assignments, an explicit INSTRUCTOR type and TEACHES relationship describe those facts more clearly.

First identify instructors independently. Then associate each section with its instructor. Keep Name on INSTRUCTOR rather than repeating it as the identity of the teacher for every section.

Prediction: Do two sections with teacher name Morgan prove that the same instructor teaches both?

Replace an entity reference with a relationship

Replace an entity reference with a relationship

Original teaching illustration | Read with the worked example

Initial description

Section	Teacher name
DB101 / 1	Morgan
DB101 / 2	Morgan

Refined facts

Instructor	Section taught
I1	DB101 / 1
I1	DB101 / 2
I2	CS102 / 1

Name remains an attribute of INSTRUCTOR.

Use INSTRUCTOR and TEACHES when instructor identity and name are needed. A teacher name alone need not identify a person.

Use INSTRUCTOR and TEACHES when instructor identity and name are needed. A teacher name alone need not identify a person.

Read the Diagram

No. Name equality is not a guaranteed identity rule. The refined example explicitly assigns I1 to both DB101 sections; I2 teaches CS102 / 1. Those assignments, not the repeated name alone, establish who teaches what.

Use singular nouns for types, descriptive verbs for relationships, and role names where needed. A noun in a requirement is a candidate for analysis, not an automatic entity type. A location may remain an attribute if only its label is needed; it may need its own type when independent room facts and relationships matter.

Do not invent a minimum from an incomplete statement such as "instructors teach sections." Ask whether instructors may teach none and whether a section must already have an instructor.

Practice: For "each course has a room," write two unanswered design questions about sections and room sharing. Then state the extra information that would justify treating ROOM as an entity rather than a simple label.

14. A Fact That Needs Three Participants

A **ternary relationship** has three participating roles. APPROVES records that a particular instructor approved a particular student for a particular course.

The complete recorded approval set is:

Record	Student	Instructor	Course
1	S101	I1	DB101
2	S101	I2	CS102
3	S102	I1	CS102

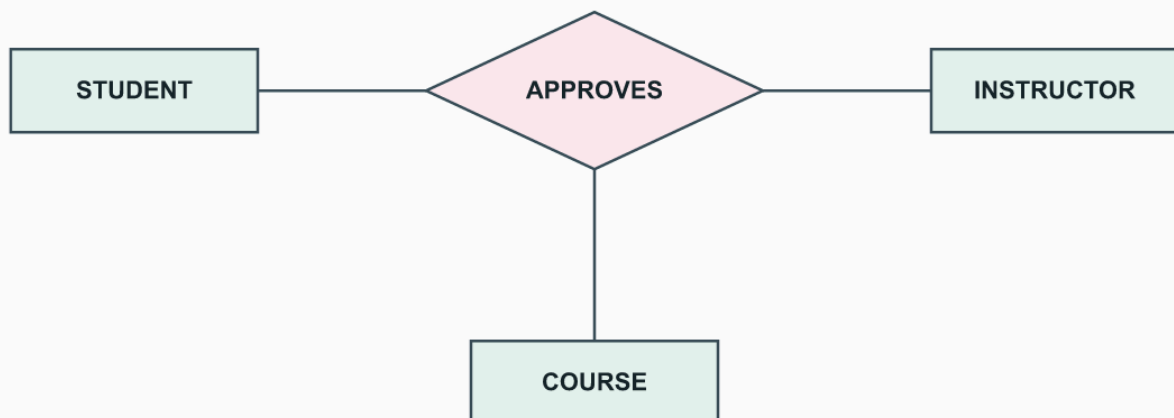
Do not assume approval merely means that the instructor has met the student or knows the course. Those would be different facts.

Prediction: The pairs S101-I1, S101-CS102, and I1-CS102 each occur somewhere in the records. Does that prove approval (S101, I1, CS102) is recorded?

One approval names a student, instructor, and course

One approval names a student, instructor, and course

Original synthetic campus example



APPROVES is ternary. Each line connects a participating type to the same diamond; it is not a sequence.

APPROVES is ternary. Each line connects a participating type to the same diamond; it is not a sequence.

Three pairwise facts do not prove a three-way fact

Three pairwise facts do not prove a three-way fact

Original teaching illustration | Read with the worked example

Recorded approvals

Student	Instructor	Course
S101	I1	DB101
S101	I2	CS102
S102	I1	CS102

Pairs for the missing triple

Pair	Seen in record
S101 + I1	1
S101 + CS102	2
I1 + CS102	3

All three pairs for (S101, I1, CS102) appear, but that approval is absent from the supplied records.

All three pairs for (S101, I1, CS102) appear, but that approval is absent from the supplied records.

Read the Diagrams

Record 1 supplies S101-I1. Record 2 supplies S101-CS102. Record 3 supplies I1-CS102. No record supplies all three together. Joining the three pairwise observations in your reasoning can therefore suggest a fact missing from the original set.

This counterexample shows why these pairs are insufficient for this case. It does not claim that ternary relationships can never be represented another way: additional constraints or a different conceptual representation may allow it.

The diamond has three undirected connections. It describes a simultaneous fact, not a flow from student to instructor to course. Repeated approvals for the same triple at different times would require an extended model.

Practice: Add exactly the missing approval to the complete recorded set. Check whether any new pairwise combination appears. Compare the two full sets to explain what the pairwise information cannot distinguish. No SQL join syntax is needed for this reasoning.

14.1. Section 3.9.2: Fix Two Participants before Reading a 1

Keep the same three recorded approvals, but introduce this explicit rule for a new variant: **for each student-course pair, at most one instructor may approve**. Put 1 on the INSTRUCTOR connection and M and N on the other two connections.

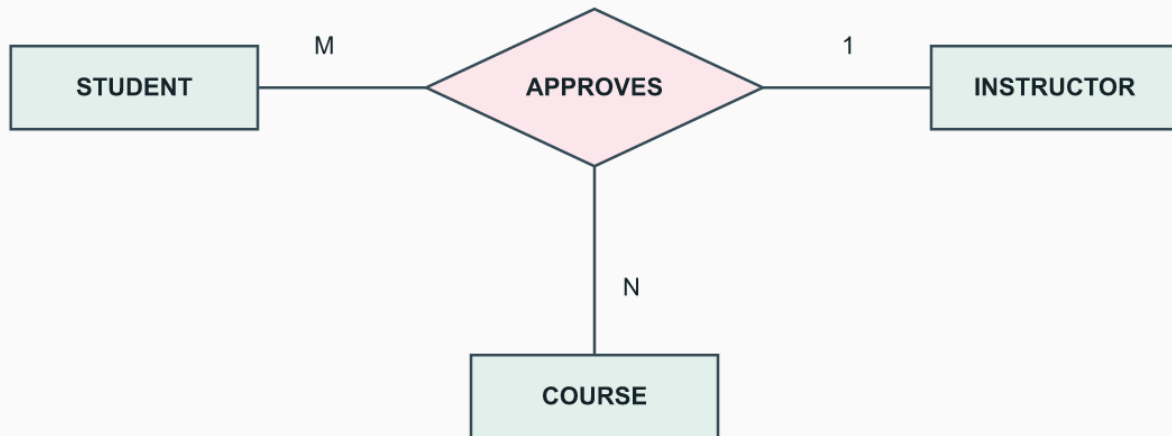
For a ternary relationship, the 1 means that fixing the other two participants allows at most one entity on this side. It does not mean that an instructor can appear in only one approval overall. The student-course pair identifies an approval instance under this rule. If all three labels are M or N, all three participants are needed in general; additional stated rules could change that.

Prediction: Does I1 appearing in records 1 and 3 violate the 1 beside INSTRUCTOR? Would adding (S101, I2, DB101) violate it?

Fix student and course: at most one instructor

Fix student and course: at most one instructor

Original synthetic campus example



The 1 limits the instructor for a fixed student-course pair, not all approvals involving that instructor.

The 1 limits the instructor for a fixed student-course pair, not all approvals involving that instructor.

Read the Ternary Maximum Diagram

I1 can approve S101 for DB101 and S102 for CS102: the student-course pairs differ. The added record would give the fixed pair (S101, DB101) both I1 and I2. That violates the rule, even though each of the three entities already exists.

For a relationship with n participating roles, the same reading fixes the other $n-1$ participants. Do not read the 1 as a limit on the total number of relationship instances containing that instructor.

Practice: Add (S103, I1, DB101) instead. Name the pair you must fix before deciding whether the maximum rule is violated.

14.2. Section 3.9.2: Count All Participations with Min-Max

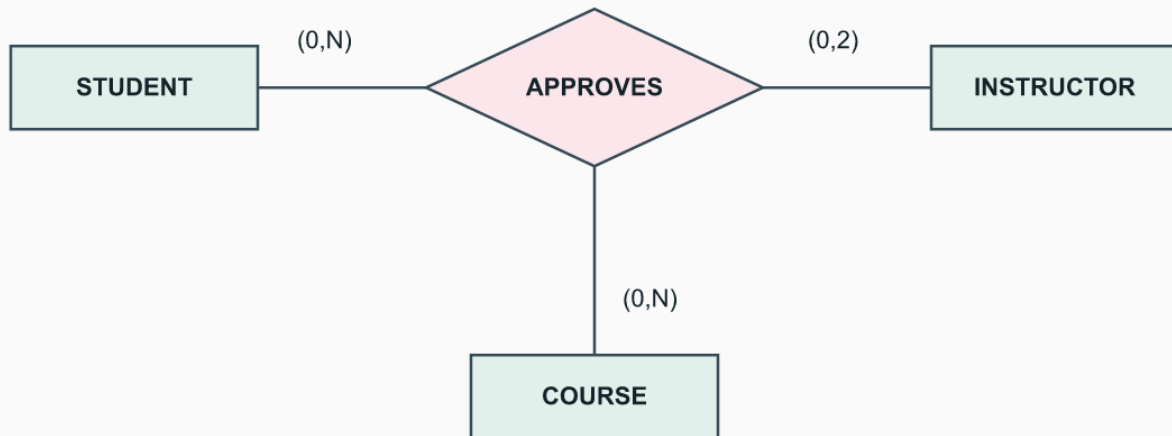
Now impose a different rule for comparison: each instructor may participate in zero, one, or two recorded approvals. Put (0,2) beside INSTRUCTOR. Student and course participation is (0,N). These are illustrative limits, not rules of the earlier core model or institutional policy.

Prediction: How many approvals involve I1 in the original three records? Would adding (S103, I1, DB101) satisfy the instructor's (0,2) constraint?

Count every approval involving one instructor

Count every approval involving one instructor

Original synthetic campus example



(0,2) permits zero to two approval instances per instructor; it does not enforce one instructor per student-course pair.

(0,2) permits zero to two approval instances per instructor; it does not enforce one instructor per student-course pair.

Read the Ternary Participation Diagram

I1 participates twice, and I2 participates once. The proposed addition raises I1's count to three, so it violates (0,2). It does not reuse a student-course pair, so it passes the previous variant's pair-based maximum rule.

The two notations are **not interchangeable for ternary relationships**. One constrains which instructor may complete a fixed student-course pair. The other limits the number of approval instances involving one instructor. Unlike the binary translation in Section 9, replacing one notation with the other would lose a requirement.

Practice: Add (S101, I2, DB101) to the original set and count I2's participations. Then test the fixed student-course pair separately. Keep the original inputs the same for both checks.

14.3. Use Both Checks When Both Rules Are Required

Assume both rules now apply: at most one instructor per student-course pair, and at most two approvals per instructor. Each proposed addition below is tested separately against the original three records, not after the preceding addition.

Proposed addition	Check the fixed student-course pair	Check total approvals per instructor
S101, I2, DB101	Does the pair already have another instructor?	How many approvals would I2 have?
S103, I1, DB101	Does the pair already have another instructor?	How many approvals would I1 have?
S103, I2, DB101	Does the pair already have another instructor?	How many approvals would I2 have?

Prediction: Which addition passes both checks?

Test the same proposed addition against two different rules

Test the same proposed addition against two different rules

Original teaching illustration | Read with the worked example

Pair rule

Addition	One instructor per pair?
S101, I2, DB101	No
S103, I1, DB101	Yes
S103, I2, DB101	Yes

Participation rule

Instructor's new count	At most two?
I2: 2	Yes
I1: 3	No
I2: 2	Yes

Each row starts from the original three approvals. Only (S103, I2, DB101) passes both rules.

Each row starts from the original three approvals. Only (S103, I2, DB101) passes both rules.

Read the Two-Check Table

The first fails the pair rule but leaves I2 with two approvals. The second uses a new pair but gives I1 three approvals. The third uses a new pair and leaves I2 with two approvals, so it passes both.

A pair-based restriction and a participation bound must both be retained when both are requirements. We show the two diagrams separately and state their conjunction explicitly, so the labels cannot be mistaken for one notation. For higher-degree relationships, do not infer the identifying combination merely from total-participation counts.

Practice: Propose another addition that passes both rules and one that fails only the total-count rule. Record the fixed pair and the before/after count. This remains paper reasoning; no relational mapping is required.

15. Assemble the Campus Diagram

Build the core diagram in this order:

1. Draw STUDENT, INSTRUCTOR, COURSE, and weak SECTION.
2. Add their keys, the partial key of SECTION, and STUDENT's multivalued Phone.

3. Connect COURSE and SECTION with identifying HAS_SECTION.
4. Connect INSTRUCTOR and SECTION with TEACHES.
5. Connect STUDENT and SECTION with ENROLLS_IN, then add Grade to the relationship.
6. Connect STUDENT, INSTRUCTOR, and COURSE to the single APPROVES diamond.
7. Check each binary relationship's maximum counts and participation separately.

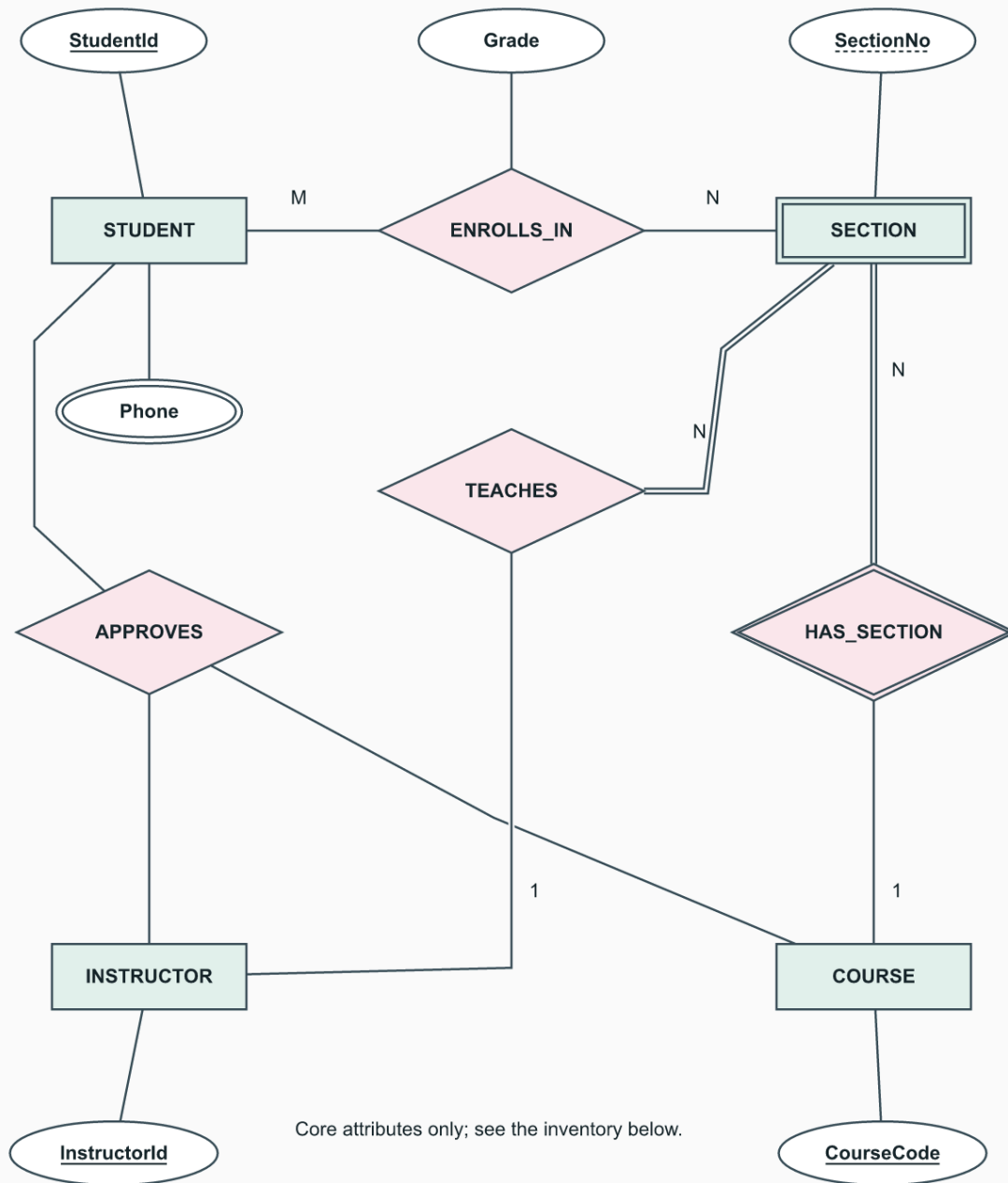
This is the deliberately small **core** model from Section 1. The expanded profile, room, card, and mentoring examples illustrate other concepts; they are not silently added to the core requirements.

Prediction: Which two relationships require total participation by SECTION? Does the total-participation line in either one require every student to enroll?

The core campus ER schema for Fall 2026

The core campus ER schema for Fall 2026

Original synthetic campus example



Maximum ratios and double participation lines describe the binary relationships. APPROVES records three-way facts.

Maximum ratios and double participation lines describe the binary relationships. APPROVES records three-way facts.

Read the Diagram

SECTION participates totally in HAS_SECTION and TEACHES: it must have both an owner course and an instructor. Those constraints do not apply to STUDENT's participation in ENROLLS_IN. S103 may still have no enrollment.

Here the diagram returns to maximum ratios and single/double participation lines, not the alternative min-max convention. Bent lines have the same meaning as straight ones. A line crossing is not a relationship or a junction; only named endpoints define participation.

The complete core attribute inventory is intentionally small:

Type or relationship	Attributes in the core
STUDENT	StudentId, Phone (multivalued)
INSTRUCTOR	InstructorId
COURSE	CourseCode
SECTION	SectionNo (partial key)
ENROLLS_IN	Grade, unknown until available

Read this diagram as a schema for possible legal states, not as a picture of only today's sample records. The sample data demonstrate consequences; they do not establish the requirements.

Practice: Redraw the diagram from Section 1 without looking at the figure. For each binary relationship, write two sentences, one from each participating type's point of view. Test an empty course, an unassigned section, a student with no enrollment, and two students sharing one section. Cite the exact requirement for each decision. Retain the ternary approval fact as a three-part fact.

16. Section 3.10: A UNIVERSITY Database

Now move from the small one-term teaching model to the book's larger UNIVERSITY case (pp.122-124, Figure 3.20). **These are different requirements**, not a correction that makes every SECTION weak or every SECTION strong.

This case includes several years of sections, a globally unique SecId, and at least five students in each section. The earlier core allows an empty section and identifies it through a course within one term. Keep the two sets of rules separate.

Prediction: Which earlier weak-entity symbol must change when SecId alone identifies every section across all stored terms?

Read the Change in Requirements

SECTION is regular here: SecId is its own key. Draw a single rectangle and an ordinary SECS relationship to COURSE. SECTION still requires one course and one instructor. Required relationships do not remove its independent key.

Practice: State two other rules that differ from the earlier core, and show a small state allowed by one model but forbidden by the other.

16.1. Identify the Entities and Attributes

Begin with six types. The inventory below paraphrases the source requirements; identifiers and attribute names match the book where useful for comparison.

Type	Identifying attributes	Other information
COLLEGE	CName	COffice, CPhone
DEPT	DCode and DName are independent keys in the requirements	DOffice, DPhone
COURSE	CCode and CoName are independently unique in the requirements	Level, Credits, CDesc
INSTRUCTOR	Id	IName, IOffice, IPhone, Rank
STUDENT	SId	SName(FName, MName, LName), Addr, Phone, Major, DOB
SECTION	SecId, globally unique	SecNo, Sem, Year, CRoom(Bldg, RoomNo), DaysTime

SName and CRoom are composite attributes. Grade belongs to TAKES, not STUDENT alone. CStartDate belongs to CHAIR, because it records when a particular instructor began chairing a department. Level describes the course's study level: 1 through 4 for undergraduate years, 5 for master's level, and 6 for doctoral level. These are the book's example codes, not a new rule for this course.

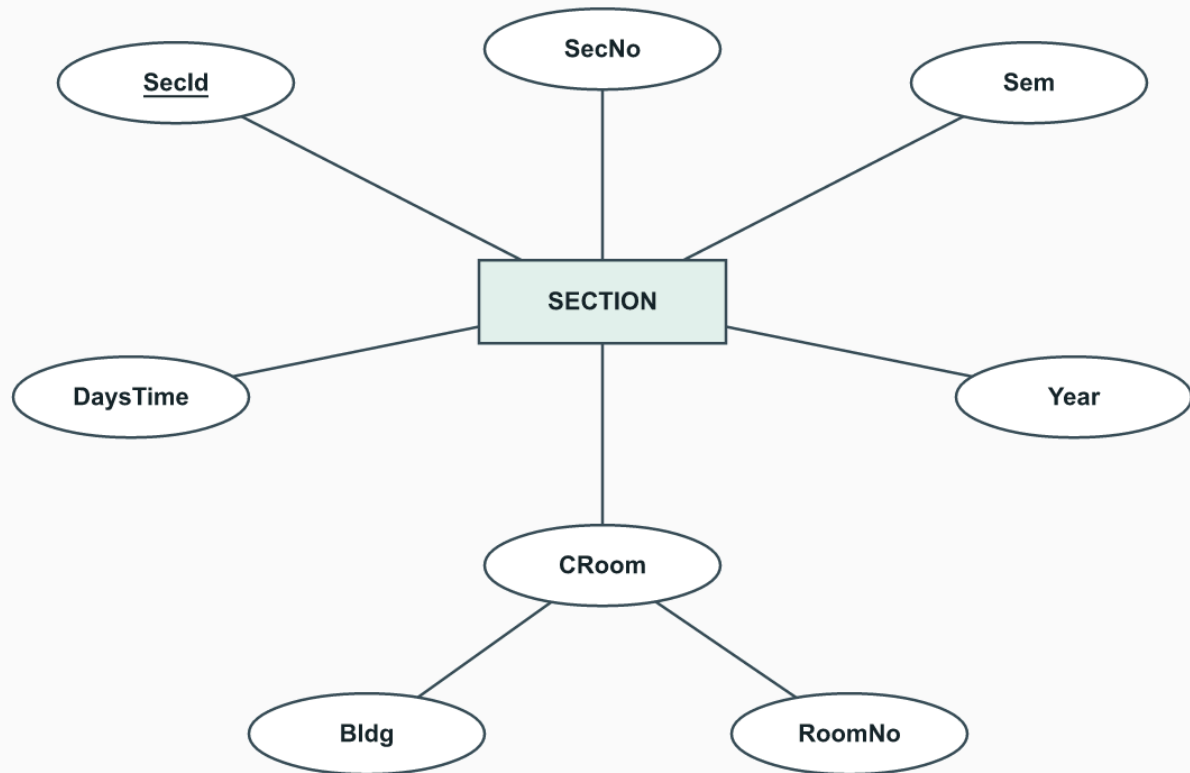
Figure 3.20 underlines both CCode and CoName. Each independently identifies a course under the stated requirements. They are two separate keys, not one composite key consisting of both attributes.

Prediction: A section is in building A, room 201. Which two components belong under CRoom? Does SecNo become a global key because SecId is one?

In Section 3.10, SECTION has a globally unique SecId

In Section 3.10, SECTION has a globally unique SecId

Original teaching layout | Textbook Section 3.10



SecId is a full key. CRoom has Building and RoomNo components; SecNo is not a global identifier.

SecId is a full key. CRoom has Building and RoomNo components; SecNo is not a global identifier.

Read the Section Attributes

CRoom contains Bldg and RoomNo. The SecId oval is underlined; SecNo is not. Two sections can use SecNo 1 in different courses or terms while having different SecId values. Entity attributes describe a section; the related course and instructor are connected through relationships rather than extra ER key ovals.

Practice: Draw STUDENT with its composite SName and its key SId. Then attach Grade to TAKES and explain the two participants needed to interpret it.

16.2. Build the Organization and Staff Relationships

Read Figure 3.20 in small parts. The next diagram uses min-max notation beside the entity whose participation is counted. Repeated rectangles with the same name denote the same entity type across the panels; they are not new types.

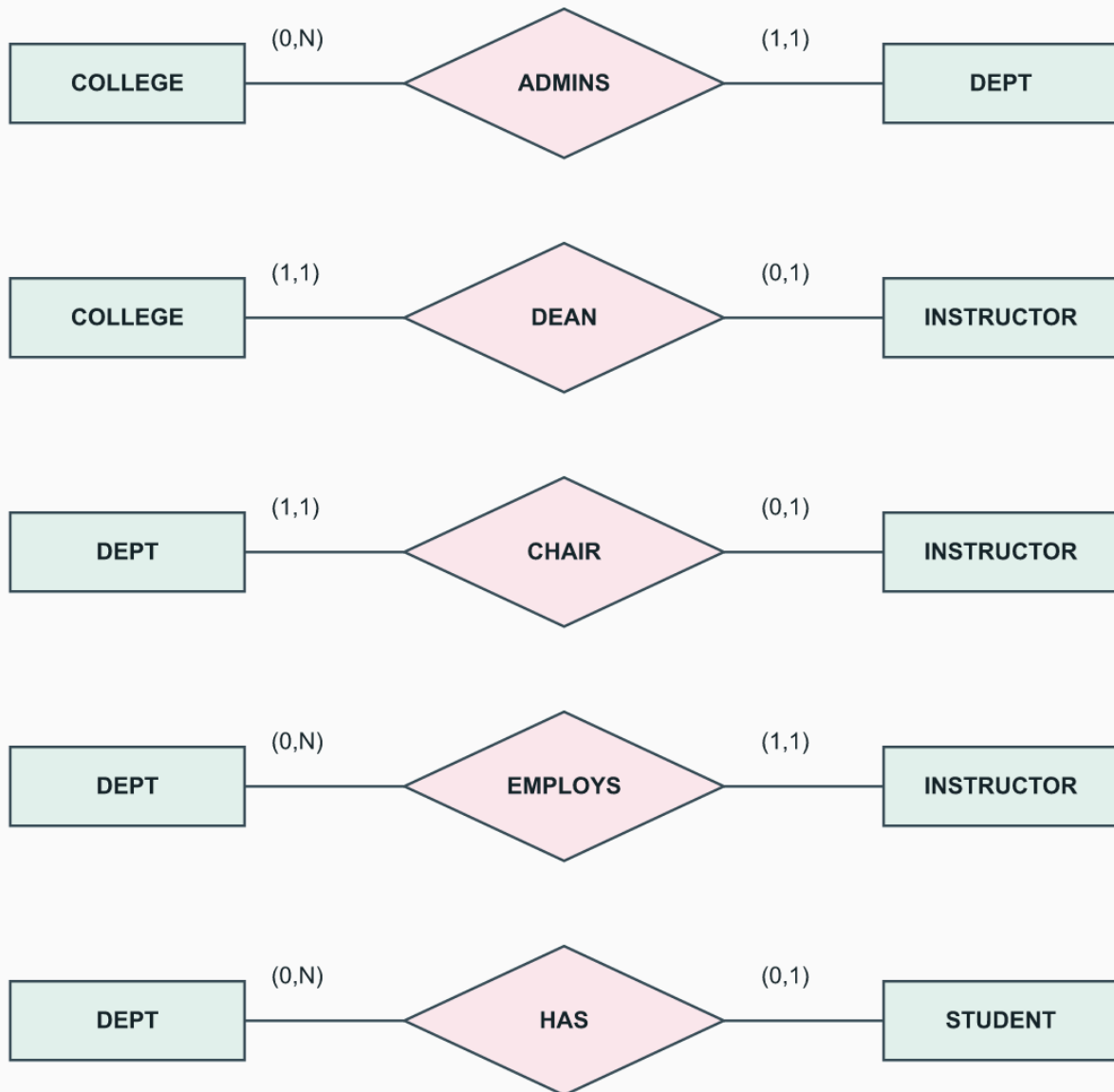
Relationship	Left participant and count	Right participant and count
ADMINS	COLLEGE (0,N)	DEPT (1,1)
DEAN	COLLEGE (1,1)	INSTRUCTOR (0,1)
CHAIR	DEPT (1,1)	INSTRUCTOR (0,1)
EMPLOYS	DEPT (0,N)	INSTRUCTOR (1,1)
HAS	DEPT (0,N)	STUDENT (0,1) in the figure

Prediction: Does an instructor who chairs a department automatically have to be that department's employee according to these connections alone?

Read the organization one relationship at a time

Read the organization one relationship at a time

Original teaching layout | Textbook Section 3.10



Counts follow Figure 3.20. HAS shows STUDENT (0,1); the prose instead requires (1,1). Keep that discrepancy explicit.

Counts follow Figure 3.20. HAS shows STUDENT (0,1); the prose instead requires (1,1). Keep that discrepancy explicit.

Read the Organization Diagrams

No. CHAIR and EMPLOYS are different relationships. Matching their department participants requires an additional rule. Likewise, dean membership in the college's departments is not established merely by having these relationship types.

For an invented snapshot, let college C-A administer departments D1 and D2. I1 is dean of C-A, I1 chairs D1, I2 chairs D2, and each works in the department they chair. A third instructor I3 may work in D1 without being a dean or chair. These facts satisfy the displayed maxima; they do not prove those rules universally.

There is a source inconsistency worth reading carefully: the prose on p.123 assigns each student to one primary department, but HAS in Figure 3.20 shows (0,1) beside STUDENT. Our diagram faithfully labels the figure's (0,1); the prose version would instead use (1,1). Do not silently merge the two. A student with no department passes the figure's version and fails the prose version. For this exercise, identify which version you are checking.

Practice: Give I3 no department. Test EMPLOYS without confusing it with CHAIR. Then redraw HAS for the prose requirement and state the one changed minimum. Attach a start date to a CHAIR instance, not to all of an instructor's roles.

16.3. Connect Courses, Sections, Teachers, and Students

The remaining relationships connect teaching and enrollment. Keep Figure 3.20's min-max convention; the counts belong beside the participating entity.

Relationship	Left participant and count	Right participant and count
OFFERS	DEPT (0,N)	COURSE (1,1)
SECS	COURSE (0,N)	SECTION (1,1)
TEACHES	INSTRUCTOR (0,N)	SECTION (1,1)
TAKES	STUDENT (0,N)	SECTION (5,N)

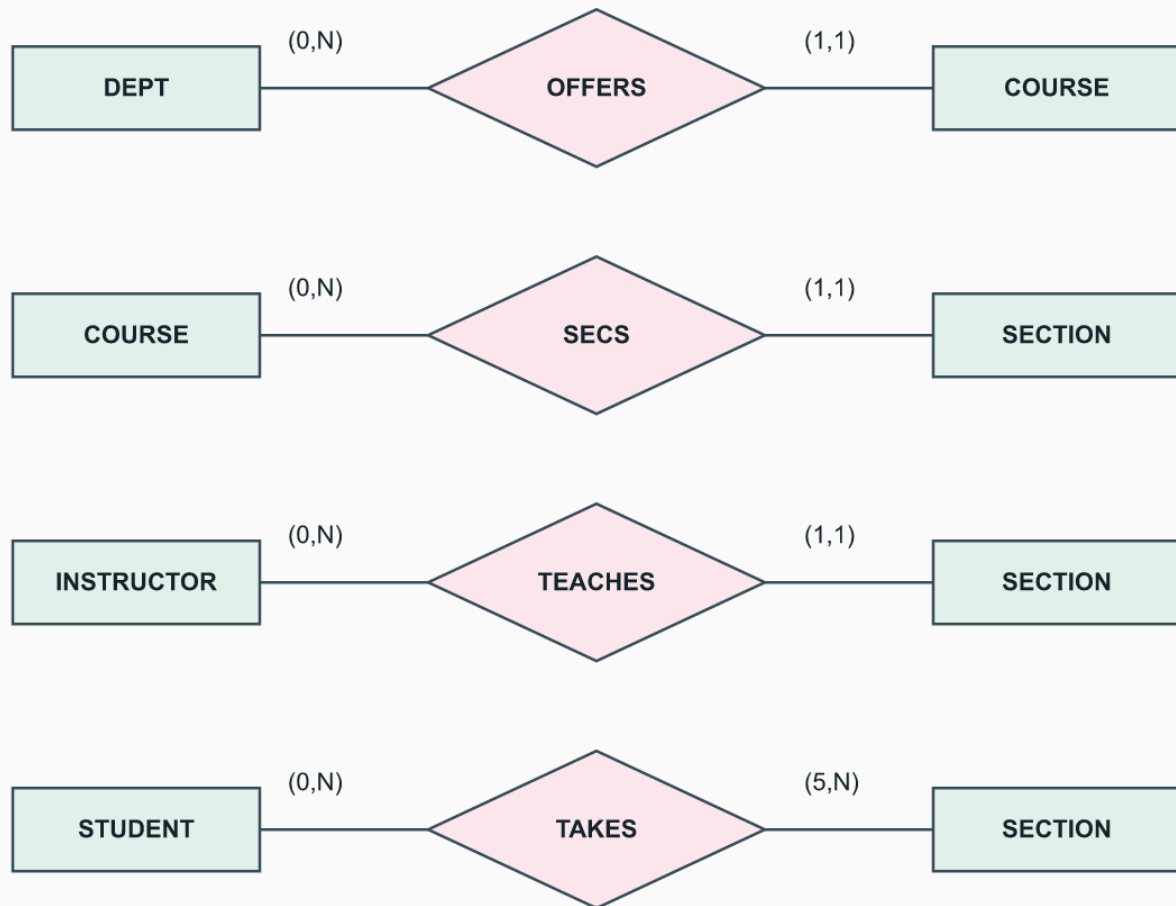
For a synthetic example, Q101 is one section of DB101, taught by I1. Students U1 through U5 take Q101. A grade can be absent until it is available. These are five distinct enrollment facts, not five duplicates of one student's fact.

Prediction: If U5 leaves and nobody replaces U5, which minimum is violated? Does an instructor with no sections violate TEACHES?

Link courses and sections without making SECTION weak

Link courses and sections without making SECTION weak

Original teaching layout | Textbook Section 3.10



Each SECTION needs a course, an instructor, and at least five students. A COURSE may have no section.

Each SECTION needs a course, an instructor, and at least five students. A COURSE may have no section.

Read the Teaching Diagrams

Q101 would have four students and violate SECTION's (5,N) participation in TAKES. An instructor with no teaching assignments satisfies (0,N). Neither condition changes SECTION's own key SecId.

Together, the organization panels and these teaching panels cover all nine relationship types in Figure 3.20. Combine repeated entity types into six rectangles to assemble the complete design. Add the attribute inventory, Grade on TAKES, and CStartDate on CHAIR. These original layouts explain the source without reproducing its page image.

Practice: Compare the state with four students against the earlier core's ENROLLS_IN rule. Explain why it is legal there but not in this UNIVERSITY example. Then reconstruct OFFERS, SECS, and TEACHES while keeping their different participants.

16.4. Some Uniqueness Rules Need More Than a SecId Oval

SecId distinguishes section objects, but different IDs alone do not prevent duplicate section numbers or scheduling conflicts.

Under the additional rules discussed on p.123, check these combinations:

Combination that must not repeat	What a repetition would mean
Course, Sem, Year, SecNo	Two sections have the same local number in the same course and term.
Sem, Year, CRoom, DaysTime	Two sections occupy the same room at the same scheduled time.
Sem, Year, instructor, DaysTime	One instructor teaches two sections at the same scheduled time.

Course and instructor here identify related entities; they are not newly added standalone SECTION attributes. The instructor rule assumes combined sections are not permitted, as the source explicitly qualifies.

Use a new isolated proposed pair for each check:

Case	SecId values	Facts shared by the two sections	Result to predict
A	Q101, Q102	DB101, Fall, 2026, SecNo 1	Local section-number conflict?
B	Q201, Q202	Fall, 2026, room A/201, Thu-P5	Room conflict?
C	Q301, Q302	Fall, 2026, instructor I1, Thu-P5	Instructor conflict?

The IDs are different in every case. Unlisted fields can be chosen differently so each example isolates the named rule.

Prediction: Does unique SecId alone make any of these pairs acceptable?

Different section IDs do not prevent other conflicts

Different section IDs do not prevent other conflicts

Original teaching illustration | Read with the worked example

Proposed sections

Case	SecId pair	Repeated combination
A	Q101 / Q102	Course, term, SecNo
B	Q201 / Q202	Term, room, time
C	Q301 / Q302	Term, instructor, time

Reason to reject

Case	Conflict
A	Duplicate local number
B	Room already occupied
C	Instructor already teaching

Each pair has different SecId values but violates its named extra rule. The three cases are checked independently.

Each pair has different SecId values but violates its named extra rule. The three cases are checked independently.

Read the Conflict Table

All three proposed pairs violate their named additional rule despite different SecId values. In case A, changing the second SecNo to 2 removes the named number conflict. In B, a different nonoverlapping time removes the named room conflict. In C, a different teacher or nonoverlapping time removes the named instructor conflict. Other requirements must still be checked.

Exact equality of DaysTime detects identical stored time patterns. If arbitrary overlapping intervals are allowed, different labels could still overlap; the simple combination check is then insufficient. This is a limitation of the equality illustration, not a claim that the book provides a complete scheduler.

Practice: Move the second section in case A to Spring 2027 and check the course-term-number combination again. Then give two differently named time intervals that overlap and explain why equality alone misses that conflict.

17. Compare Designs and Practice

Use the same requirements for every group's design:

- A section belongs to exactly one course.
- SectionNo is unique only within its course.
- A section has exactly one instructor; an instructor may teach none or several.
- A student may enroll in none or several sections.
- A section may have none or several students.

Each group draws COURSE, SECTION, INSTRUCTOR, and STUDENT with these relationships and constraints, and explains its identification choice. Compare the same facts, not artistic style or the number of colors used. A diagram may use either of the two taught constraint notations if it uses that notation consistently.

After the group responses are fixed, compare the anonymously displayed responses individually. For each response, test DB101 / 1 alongside CS102 / 1, an instructor with no assignment, and a section with two instructors. Use those checks to explain your ordering of the responses. The instructor will discuss the technical reasons before you revise your own drawing. Peer ranking does not determine grades.

Keep your own drawing and one checked prediction as learning notes. Include one prediction, the observed result, and either your correction or an explanation of why your prediction was correct. Submit only the work the instructor assigns.

For further individual practice, change one requirement at a time:

Change	What to inspect
Team teaching is allowed.	The maximum instructor count per section
Every student must enroll somewhere.	The minimum participation of STUDENT in ENROLLS_IN
Several terms must be retained.	Whether the current section identity still distinguishes all sections
One globally unique SectionId is supplied.	Whether identification still depends on the owner

Do not treat a permitted state as a required state, and do not change unrelated rules to make a proposed diagram work.

Chapter Summary and Next Chapter

An entity type describes objects; an entity set records the objects currently present. Attributes describe properties. Keys identify entities under rules that apply to all legal states.

Relationships connect participating entities. Maximum cardinality and minimum participation answer different questions. A relationship attribute describes a particular connection. A weak entity needs its owner's identity, and a ternary fact may contain information lost by considering pairs alone. For ternary relationships, fixing other participants and counting total participation are different checks. The UNIVERSITY case illustrates how changed identification, minimum-enrollment, and multi-term requirements change a design.

Ch4 adds the **Enhanced Entity-Relationship (EER) model**. It will ask how a general type and more specialized types can share properties, and how membership rules should be represented. Keep this campus model for that discussion. Relational mapping and SQL implementation come later; they are not prerequisites for completing this chapter's drawings.

Textbook Reading

The explanations and examples use original wording and synthetic data. The conceptual definitions follow Chapter 3 of the prescribed book.

Material here	Textbook sections and printed pages
Requirements and conceptual design	3.1-3.2, pp.90-93
Entities, attributes, keys, domains	3.3, pp.93-102
Relationships, roles, constraints, relationship attributes	3.4, pp.102-108
Weak entities, nested owners, and multiple owners	3.5, p.109; 3.9.1, pp.120-121
Refinement, ER symbols, min-max notation	3.6-3.7, pp.110-115
Simple ternary relationship	3.9.1, pp.118-121
Ternary and higher-degree constraints	3.9.2, pp.121-122
UNIVERSITY requirements, diagram, and additional uniqueness rules	3.10, pp.122-124; Figure 3.20

Page numbers refer to the photographed classroom copy. Chapter and section numbers are the preferred locators when another printing has different pagination.

Detailed UML notation in Section 3.8 is not assigned here. The new UNIVERSITY walkthrough paraphrases the source and explicitly identifies its prose/figure differences; its input records and diagram layouts are original teaching examples.